

CSR-2315 Data Structures

Week 3: Worst-Case Run Time Analysis

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1. `is_even(N)`

Answer: $O(1)$

Reasoning: The function performs a fixed number of operations no matter the value of N ; N is either divisible by 2 or not.

2. `find_number(arr, number)`

Answer: $O(N)$

Reasoning: In worst case scenario, the function must check every element to ensure that the same value isn't present.

3. `constant_multiplier(arr)`

Answer: $O(N)$

Reasoning: The loop goes through every element in the array once. Since there are N elements, the number of operations grows linearly with N .

4. `arr = [2*elem for elem in arr]`

Answer: $O(N)$

Reasoning: The entire array is gone through once; the more items present the longer it will take.

5. `create_pairs(arr)`

Answer: $O(N^2)$

Reasoning: The function contains a loop inside another loop. The amount of time increases as much as N^2 .

6. `is_prime(N)`

Answer: $O(N)$

Reasoning: The loop checks every integer from 2 up to N . Therefore, the number of operations grows linearly with N , which increases time.

7. `print_halves(N)`

Answer: $O(\log N)$

Reasoning: Each iteration divides N by 2 ($N/2$). The number of times N is divided by 2 the slower the results, i.e. 12 loops.

8. `double_or_nothing(arr)`

Answer: $O(N)$

Reasoning: Two-step process, each processing the elements once.

9. Determine the Big-O notation from the exact operation count.

A. N^2

Answer: $O(N^2)$

Reasoning: The expression contains N^2 as its highest-order term; $O(N^2)$.

B. $3N + 5$

Answer: $O(N)$

Reasoning: Constants 3 and 5 are being ignored in Big-O notation. The highest-order term is N , so the run time is $O(N)$.

C. 36,000

Answer: $O(1)$

Reasoning: The number of operations is constant and not depending on N .

D. $N^3 + 2N^2 + N + 3000$

Answer: $O(N^3)$

Reasoning: The highest-order term is N^3 ; $O(N^3)$

E. $(1000N + 1) / (1000N^2)$

Answer: $O(1)$

Reasoning: As the expression gets smaller, N grows larger; not being linear, $O(1)$.

F. $N \times (N - 1) \times (N - 2) \times (N - 3) \times \dots \times 1$

Answer: $O(N!)$

Reasoning: The expression is N factorial; therefore, its run time is $O(N!)$.

G. $\Sigma(N - i)$, from $i = 0$ to N

Answer: $O(N^2)$

Reasoning: The sum produces approximately $N^2 / 2$ operations, giving $O(N^2)$.